

GeoNail System Architecture & Comprehensive Technical Manual

Smart Subterranean Anchoring, Slope Stability Monitoring & Real-Time Digital Twin System

1. Executive Summary

GeoNail is an advanced Internet of Subsurface Things (IoST) geotechnical monitoring platform designed for real-time detection of slope instability, structural micro-displacements, and seismic anomalies in high-risk mining, civil excavation, and landslide-prone environments.

The system integrates high-precision embedded sensor hardware, on-device edge anomaly processing, multi-transport telemetry (BLE, Serial UART, Wi-Fi/MQTT), and a 1:1 real-world scale 3D Web CAD & Digital Twin Visualizer built on Three.js and React.

+-----+ GEONAIL SYSTEM +-----+					
+-----+	+-----+		+-----+		
Subsurface Anchors	M5Stack Edge Node		GeoNail OS Web		
- Magnetometer Array	-> - ESP32-S3 Dual-Core	->	- 3D CAD X-Ray		
- 3-Axis Accelerom.	- Compl. Filter / EMA		- 1:1 Minefield		
- Gyro Sensor	- Real-time Edge Rules		- BLE/Serial Modem		
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2. Hardware Architecture & Mechanical Design

2.1 Mechanical Form Factor

The GeoNail housing consists of a heavy-duty industrial-grade sealed polycarbonate shaft housing the electronics core, topped with a locked protective enclosure cap and anchored deep into soil/rock strata via a telescoping magnetic spike anchor.

- **Main Enclosure Body:** Cylindrical shell containing M5Stack Core2 / ESP32-S3 MCU, OLED/IPS display, lithium-polymer power cell, and status transducer (piezo/buzzer).
- **Telescoping Spike Anchor:** Underground anchor embedded directly into stable soil layers. As soil shifts, relative displacement between the outer casing and inner spike anchor alters internal magnetic field geometry.
- **Locked Top Cap:** Moisture-sealed protective cap housing external antennas and status LEDs.

2.2 Microcontroller & Sensors

- **Microcontroller:** ESP32-S3 (Dual-Core Xtensa LX7 @ 240 MHz, 512KB SRAM, 8MB PSRAM, Wi-Fi 4 + BLE 5.0).
- **Power Management Unit (PMU):** AXP192 integrated power controller supporting battery state-of-charge tracking, dynamic voltage scaling, and solar/external power charging.
- **Inertial Measurement Unit (IMU):** MPU6050 / MPU6886 6-Axis motion sensor (3-axis Accelerometer, 3-axis Gyroscope).
- **Magnetometer / Hall Sensor Array:** Precision magnetic sensing elements configured along the inner shaft axis to resolve absolute linear displacement of the subterranean anchor rod.

3. Firmware Architecture & Edge Signal Processing

The firmware is written in C++ for the Arduino/ESP32 framework (`GeoNail_M5Stack_Firmware.ino`). It executes multi-sensor sampling at 100 Hz with real-time digital filtering and edge anomaly detection.

3.1 Sensor Fusion & Orientation Math

To compute real-time Roll, Pitch, and Yaw angles without gimbal lock, the firmware implements a Complementary Filter combining raw gyroscope angular rates with accelerometer gravity vectors.

Accelerometer Roll (ϕ_{acc}) and Pitch (θ_{acc}):

$$\phi_{acc} = \arctan2(A_y, A_z) \cdot \frac{180}{\pi}$$

$$\theta_{acc} = \arctan2(-A_x, \sqrt{A_y^2 + A_z^2}) \cdot \frac{180}{\pi}$$

Complementary Filter Equation:

$$\phi_t = \alpha \cdot (\phi_{t-1} + G_x \cdot \Delta t) + (1 - \alpha) \cdot \phi_{acc}$$

$$\theta_t = \alpha \cdot (\theta_{t-1} + G_y \cdot \Delta t) + (1 - \alpha) \cdot \theta_{acc}$$

$$\psi_t = \psi_{t-1} + G_z \cdot \Delta t \quad (\text{Yaw integration})$$

where $\alpha = 0.96$ is the filter weight tuning constant, balancing gyroscope responsiveness with accelerometer low-pass drift suppression.

3.2 Subterranean Anchor Displacement Math

The magnetic flux density B measured along the longitudinal axis varies inverse-proportionally with the insertion distance d of the magnetic spike anchor relative to the casing baseline:

$$\Delta B = B_{ref} - B_{measured}$$

$$d_{mm} = k_m \cdot \sqrt{\frac{1}{\Delta B + \epsilon}} + d_{offset}$$

- As the spike anchor shifts deeper or pulls out during soil subsidence, ΔB changes dynamically.
- The calibrated constant k_m converts gauss / tesla micro-variations into exact linear displacement in millimeters (mm) and centimeters (cm).

3.3 Edge Anomaly & Seismic Detection Engine

To eliminate environmental noise while guaranteeing immediate triggering during shear collapses or seismic shocks, the firmware utilizes dual statistical estimators:

Exponential Moving Average (EMA):

$$\text{EMA}_t = \beta \cdot X_t + (1 - \beta) \cdot \text{EMA}_{t-1}$$

where $\beta = 0.05$ smooths slow creep baseline drift.

Root Mean Square Acceleration (RMS_{vib}):

$$\text{RMS}_{\text{vib}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (A_{x,i}^2 + A_{y,i}^2 + A_{z,i}^2 - 1.0^2)}$$

Multi-Stage Trigger Thresholds:

State	Condition	Firmware Action		
NORMAL		ΔAngle	$< 2.0^\circ$, $d < 5\text{mm}$, $\text{RMS} < 0.15g$	Display regular green HUD telemetry
WARNING	$2.0^\circ \leq$	ΔAngle	$< 5.0^\circ$ OR $5\text{mm} \leq d < 15\text{mm}$	Amber LED blink, 1 Hz audio prompt
CRITICAL		ΔAngle	$\geq 5.0^\circ$ OR $d \geq 15\text{mm}$	Red LED strobe, 5 Hz piezo siren
SEISMIC_ANOMALY	$\text{RMS}_{\text{vib}} > 0.65g$ (Transient spike)	Instant emergency telemetry beacon dispatch		
ANCHOR_SLIP	$\frac{\partial d}{\partial t} > 2.5 \text{ mm/s}$	Lock memory dump, triggers immediate shear alert		

4. Telemetry Protocols & Communications

GeoNail provides continuous multi-channel telemetry streams supporting three distinct operational modes:

4.1 USB Serial UART Telemetry (115200 Baud)

Sends structured JSON packets at 20 Hz over the USB-C serial port:

```
{
  "device_id": "GEONAIL_NODE_01",
  "uptime_ms": 142850,
  "pitch": 1.45,
  "roll": -0.82,
  "yaw": 12.30,
  "displacement_mm": 3.42,
  "vibration_rms": 0.042,
  "battery_pct": 94,
  "status": "NORMAL",
  "rssi": -62
}
```

4.2 Bluetooth Low Energy (BLE) GATT Architecture

- **Device Name:** GeoNail-Node-01
- **Custom Service UUID:** 4FA10001-0000-1000-8000-00805F9B34FB
- **Characteristics:**
 - **Telemetry Data (Notify):** 4FA10002-0000-1000-8000-00805F9B34FB (Binary packed float array: [Pitch, Roll, Yaw, Displacement, Vibration]).
 - **Calibration Command (Write):** 4FA10003-0000-1000-8000-00805F9B34FB (Send 0x01 to zero IMU tilt, 0x02 to reset magnet baseline).

4.3 Wi-Fi Mesh & MQTT WebSockets

In remote deployments, nodes form a local ESP-NOW mesh relaying telemetry to a central gateway node, which publishes MQTT topics (**geonail/telemetry/node_01**) over cellular or satellite backhaul to the GeoNail OS visualizer.

5. GeoNail OS Web Visualizer Architecture

The frontend visualizer is a modern React 18 + TypeScript web application (**sih-3d-visualizer**) powered by Vite, Tailwind CSS, Lucide icons, and Three.js / React Three Fiber.

5.1 System Tab Architecture

GEONAIL OS INTERFACE			
[1. 3D CAD Viewer]	[2. MCU Dashboard]	[3. Minefield Digital Twin]	
- High-res geometry	- Real-time gauges	- 1:1 scale (800m x 600m)	
- X-Ray translucent	- Live trend graphs	- CAT 797 heavy dump trucks	
- Locked cap state	- Direct Serial/BLE	- Subterranean shear slice	
- Multi-camera system	- Alarm logs & reset	- Node deployment grid	

5.2 3D CAD Viewer Page Features

- **Full 3-Axis Reactive Model:** Real-time rotational synchronization driven by live or simulated Pitch, Roll, and Yaw angles.
- **Translucent X-Ray Mode:** Renders exterior casing with glass-like transparency ($\text{opacity} = 0.35$), exposing internal battery, M5Stack core board, magnetometer coils, and internal anchoring mechanism.
- **White Engineering Grid Aesthetics:** Crisp white grid floor ($\text{\#ffffff}$ canvas background, $\text{\#cbd5e1}$ grid lines, $\text{\#64748b}$ axis markers) for clear CAD visualization.
- **Locked Cap State:** Standardized fixed state showing locked protective cap securely attached to top shaft.
- **Multi-Preset Camera Control:** Dedicated controls for standard engineering projections:
- **Isometric View:** Position $[12, 10, 12]$, Target $[0, 0, 0]$
- **Top (Plan) View:** Position $[0, 22, 0]$, Target $[0, 0, 0]$
- **Front Elevation:** Position $[0, 0, 20]$, Target $[0, 0, 0]$
- **Side Elevation:** Position $[20, 0, 0]$, Target $[0, 0, 0]$
- **Reset Orbit:** Smoothed reset transitions via OrbitControls.

5.3 MCU Dashboard Page Features

- **Hardware Connection Modal:** Native Web Serial API & Web Bluetooth API integration allowing operators to connect directly to physical GeoNail devices from any browser.
- **Interactive Telemetry Gauges:** Real-time dial gauges for Pitch ($^{\circ}$), Roll ($^{\circ}$), Displacement (mm), and Vibration RMS (g).
- **Historical Signal Trends:** Canvas-based line graphs tracking 60-second sliding windows of ground movement.
- **System State Indicator:** Dynamic color-coded status banner with audio warning indicators.

5.4 1:1 Scale Minefield Digital Twin Page Features

- **Real-World Environment:** Models an $800\text{m} \times 600\text{m}$ open-cast mining pit complete with terraced benches (5m height steps), haulage ramps, excavation zones, and top perimeter access roads.
- **Real-World Scale Reference:** Features detailed 1:1 scale 3D models of **CAT 797 Dump Trucks** ($14.8\text{m} \times 9.8\text{m} \times 6.5\text{m}$) navigating haul roads, allowing engineers to intuitively judge scale.
- **Multi-Node Deployment Grid:** Multi-node monitoring network (**GEONAIL-01** to **GEONAIL-06**) positioned along critical slope benches. Clicking any node selects it, displaying its live telemetry HUD overlay and triggering a smooth camera glide to its coordinates.
- **Subterranean Shear Slice Visualization:** Underground cross-section plane highlighting slip surfaces, failure arcs, soil strata layers, and real-time anchor penetration depths.

6. Calibration & Operating Procedures

6.1 Initial Zero-Bias Calibration Procedure

1. **Mechanical Mounting:** Drive GeoNail outer casing into borehole until top flange rests flush against reference surface.

2. Spike Extension: Drive telescoping spike anchor into bedrock / solid substrate.

3. Firmware Zeroing:

- Hold M5Stack **Button A** for 3 seconds or trigger **Zero Calibration** in GeoNail OS Web Visualizer.
- Device records current A_x , A_y , A_z baseline offset as 0.0° pitch/roll reference.
- Device stores current magnetic flux reading B_0 as zero displacement (0.0 mm).

6.2 Troubleshooting Audio & Display Behavior

- **Audio Noise / Buzzing:** Occurs when PWM buzzer pin is left floating during WiFi transmission bursts. Solved in firmware v2.1 via active pull-down logic and low-pass filter capacitor across speaker lines.
- **Display Flicker:** Prevents display burn-in and SPI bus contention by utilizing double-buffered sprite rendering (`M5.Lcd.pushSprite()`) updated strictly at 30 FPS.

7. System Verification & Validation

The full GeoNail system architecture has undergone extensive end-to-end integration testing:

- 1. Hardware Communication:** Web Serial and BLE hardware communication modules verified at 115200 baud and 20 Hz packet rates.
- 2. Orientation Accuracy:** Roll/Pitch tracking verified within $\pm 0.2^\circ$ accuracy against physical digital clinometer.
- 3. Anchor Displacement Resolution:** Magnetometer displacement calculation validated from 0.0 mm to 50.0 mm with sub-millimeter precision.
- 4. Digital Twin Visualization:** Smooth 60 FPS 3D canvas rendering verified across CAD, MCU, and Minefield views.

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